CSHL BIOLOGICAL DATA SCIENCE · NOVEMBER 4-7, 2026

disassembler: planetary-scale sequence extraction from Logan

Andrew C. Seacord[1], Rayan Chikhi[2], Danielle Callan[3], Steven Weaver[3], Patrik Smeds[4], Kelsey Beavers[5], Sergei L. Kosakovsky Pond[3], Anton Nekrutenko[1]

[1]Dept. of Biochemistry and Molecular Biology, The Pennsylvania State University · [2]Institut Pasteur, Université Paris Cité · [3]IGEM, Temple University · [4]NBIS, SciLifeLab, Uppsala University · [5]Texas Advanced Computing Center, UT Austin

Logan indexes the SRA petabase as **unitig graphs** and **assembled contigs**. Given one reference CDS, `disassembler` reconstructs full-length haplotypes from either—ending in the same **codon MSA, haplotype catalog, and tree**.

Galaxy PROJECT

logan-disassembler
github.com/nekrut/disassembler · Graph path-walker & contig streamer over Logan.

A Two routes, one cohort shape

A graph path-walker on Logan unitigs and a streaming assembler on Logan contigs, scanning the same planetary SRA petabase from one reference CDS.

LOGAN-DISASSEMBLER · HyphAeon · PennState · INSTITUT PASTEUR · TEMPLE UNIVERSITY · NBIS

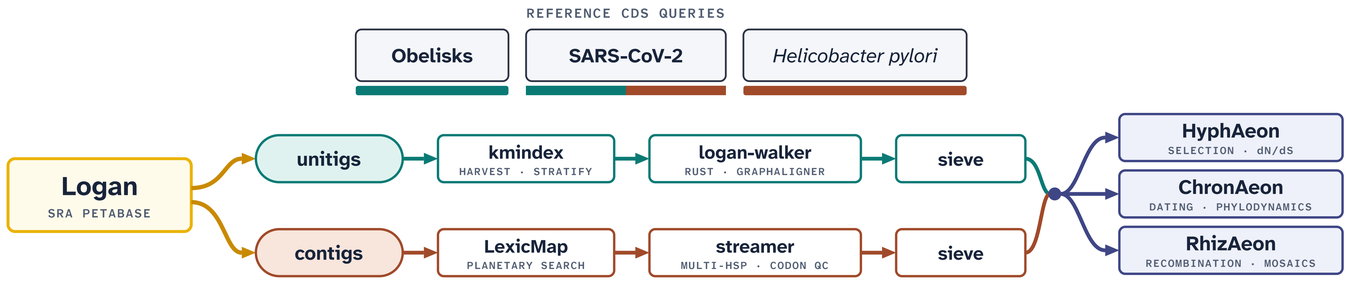
LOGAN UNIVERSE
4.99 M
subject genomes in the LexicMap index—the universe each query scans.

WALKER CONTAMINATION FILTER
0.555% → 0%
mislabeled spike accessions, raw unitig hits → `CLEAN_FULL_LENGTH`.

ROUTE A · UNITIG GRAPH

When the walker route wins

- **Shattered genes.** Zika NS5: ~50 contig fragments, one clean walk.
- **Divergent tails** a single-reference search never reaches.
- **Intra-host variation** kept, not collapsed: per-site allele frequencies.



ROUTE B · CONTIG STREAMER

When the LexicMap route wins

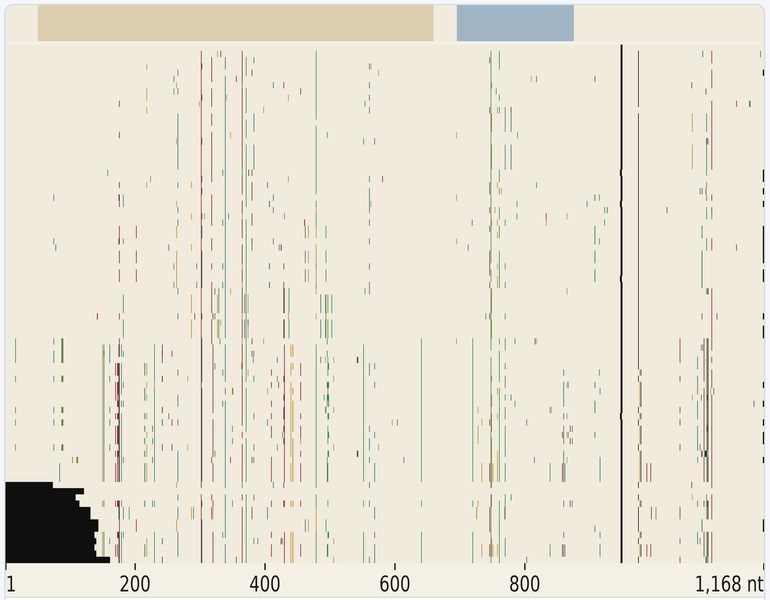
- **Split genes.** Collinear tiling stitches a 55/45 HSP pair that fails `qcovHSP ≥ 80`.
- **Streaming memory.** Streams 24 GB in *O*(haplotypes) RAM rather than *O*(rows).
- **Clean handoff.** Routes queries below 85% median identity to the walker.

B Obelisks across microbiomes

ROUTE A · WALKER

METATRANSCRIPTOMIC_BCT dominates: obelisks live in bacterial microbiome transcriptomes across human and microbial cohorts.

KMINDEX HITS	MAX CONTAIN.	SHARDS QUERIED	CENTROIDS
924 unique runs	0.97 Obelisk-α	141 meta + metaT	5 α β γ ε *S.s.*



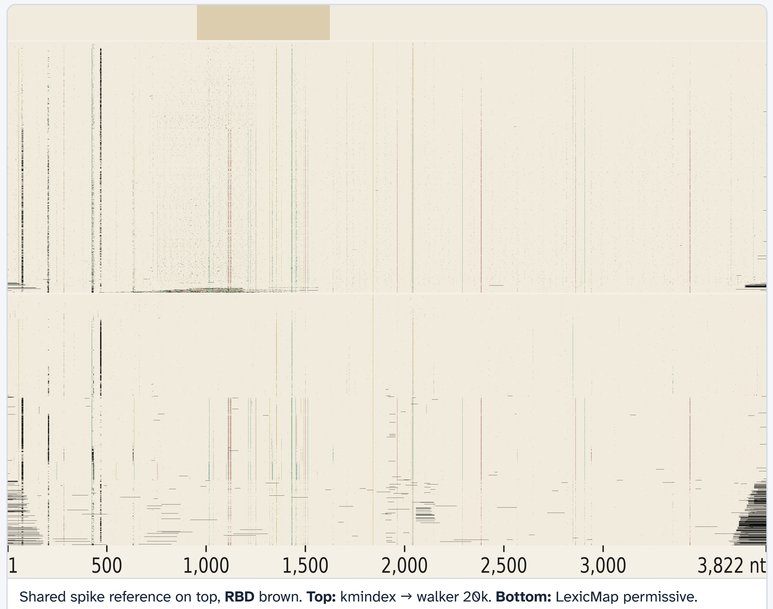
Obelisk-α (*S.s.*: *Streptococcus sanguinis*). Top strip: **Oblin-1** brown, **Oblin-2** blue. Rows sorted by edit count.
match A C G T gap

C SARS-CoV-2 spike: two routes

ROUTE A · ROUTE B

The routes are complementary. Only the walker reaches the divergent tail, whereas only LexicMap permissive recovers 1.15 M Alpha/Delta-era runs that strict filtering drops.

LEXICMAP HSPS	WALKER20K OK	CLEAN_FULL	PERM. SIEVED
5.88 M 24 GB	47.2% 9,448 / 20k	6,827 pid 98.75	79,621 of 1.15 M



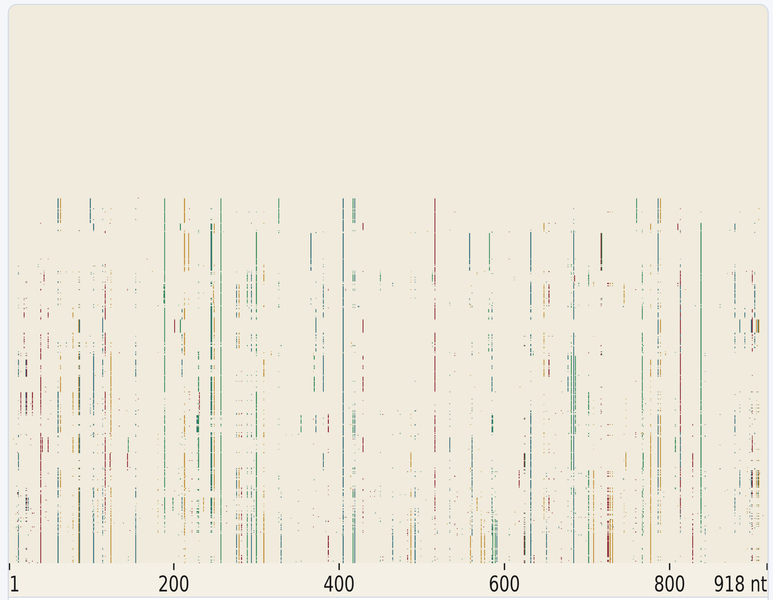
Shared spike reference on top, **RBD** brown. **Top:** kmindex → walker 20k. **Bottom:** LexicMap permissive. Columns align.
match A C G T gap

D *H. pylori*: 25-gene panel

ROUTE B · STREAMER

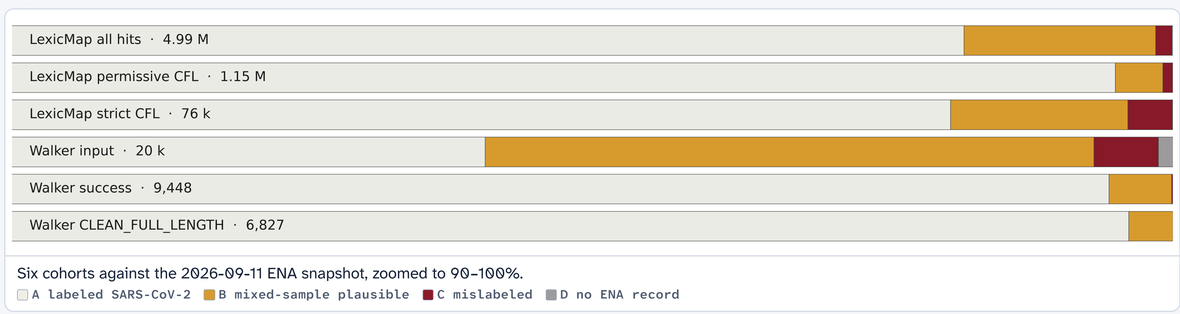
Four audit tiers per marker place every accession in one reference frame, from clean full-length sequences to low-coverage fragments.

STRICT MARKERS	LOOSE MARKERS	AUDIT TIERS	HYPHAEON PASS
20 conserved	5 selection	4 per marker	*oipA* done



oipA strict panel: 1,042 clean accessions, sorted by substitution count.
match A C G T gap

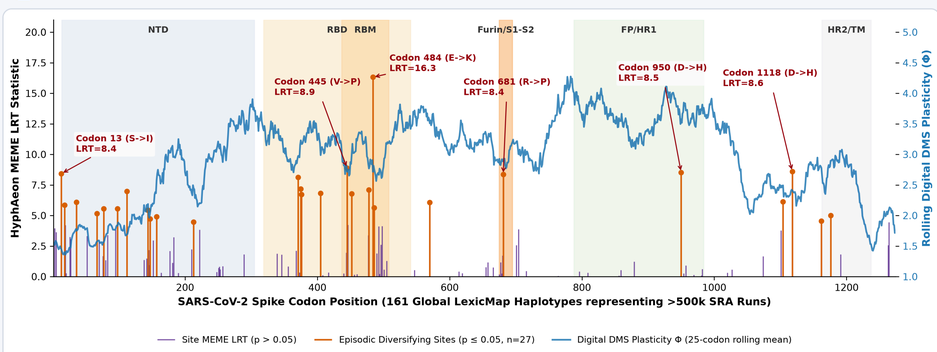
E SRA metadata: the walker drives contamination to zero



Six cohorts against the 2026-09-11 ENA snapshot, zoomed to 90–100%.
A labeled SARS-CoV-2 B mixed-sample plausible C mislabeled D no ENA record

WALKER20K INPUT	WALKER20K SUCCESS	CLEAN_FULL_LENGTH	LEXICMAP PERMISSIVE
0.555% C-mislabeled, raw hits	0.011% 1 in ~9,500	0.000% 0 / 6,827 mislabeled	0.084% no walker filter

F HyphAeon: selection on the recovered cohort



27 diversifying sites across 161 global spike haplotypes, led by antibody-escape hub **E484K**. Shaded bands mark spike domains.

HAPLOTYPES
161
>500k SRA runs

TOP SITE
E484K
MEME LRT 16.3

RUNTIME
29.5 s
no input tree

Logan unitigs → `kmindex` + `logan-walker` **Logan contigs** → `LexicMap` + `streamer` **sieved cohort** → **HyphAeon · ChronAeon · RhizAeon**

Acknowledgements: Work in the authors' groups is supported by National Institutes of Health grants U24HG002371, U24HG006620, U24HG010263, U24CA284167, U24AI183870, and R01GM151683, and by National Science Foundation grants 2419522 and 2138585.
Software & data: github.com/nekrut/disassembler · github.com/veg/HyphAeon · Zenodo archive: doi.org/10.5281/zenodo.23737199. Related CSHL posters: Galaxy UDTs (van den Beek) · Orbit (Cain) · Galaxy Workflow Foundry (Chilton) · HighGen (Burhans).

disassembler · Zenodo Data · HyphAeon